Pre-Experiment Overview: Fine Structure of Two-Electron Atom He

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Purpose of the Experiment

According to the lab manual, the main objective of this lab is to investigate the optical emission spectrum of the two-electron helium atom (He-I). By dispersing the emitted light through a high-resolution optical spectrometer equipped with a diffraction grating, we measure the diffraction angles for each visible spectral line, determine their precise wavelengths, identify singlet and triplet state transitions, and construct the spectrometer calibration curve.

Importance and Significance

Studying helium is a crucial bridge in quantum mechanics: while hydrogen is a solvable one-electron system, helium introduces multi-electron electrostatic repulsion and quantum mechanical electron exchange symmetry. Because electrons are indistinguishable fermions, the total wavefunction must be anti-symmetric under particle exchange, giving rise to two distinct term systems: parahelium with antiparallel spins (singlet states, total spin $S = 0$) and orthohelium with parallel spins (triplet states, total spin $S = 1$). Because dipole selection rules strictly forbid transitions between different spin multiplicities ($\Delta S = 0$), helium essentially behaves optically as two separate, non-combining systems.

In the lab setup, a helium discharge tube emits bright spectral lines spanning red, yellow, green, blue, and purple. A spectrometer consisting of a collimator, prism/grating table, and telescope allows us to measure left and right diffraction angles θ_{left} and θ_{right} for orders $m = 1$ and $m = 2$. Using the grating relation $m\lambda = d \sin\theta_{\text{ave}}$, we calculate the experimental wavelengths of individual transitions (such as 706.5 nm, 587.6 nm, 501.5 nm, 492.2 nm, 471.3 nm, and 447.1 nm). Plotting $\sin\theta_{\text{ave}}$ against wavelength produces an accurate linear calibration curve and allows us to verify Russell-Saunders (L-S) coupling and the fine-structure term diagram of multi-electron atoms.